VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

WINTER SEMESTER

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

LAB ASSESSMENT - 4

REG NO : 23MIS0051

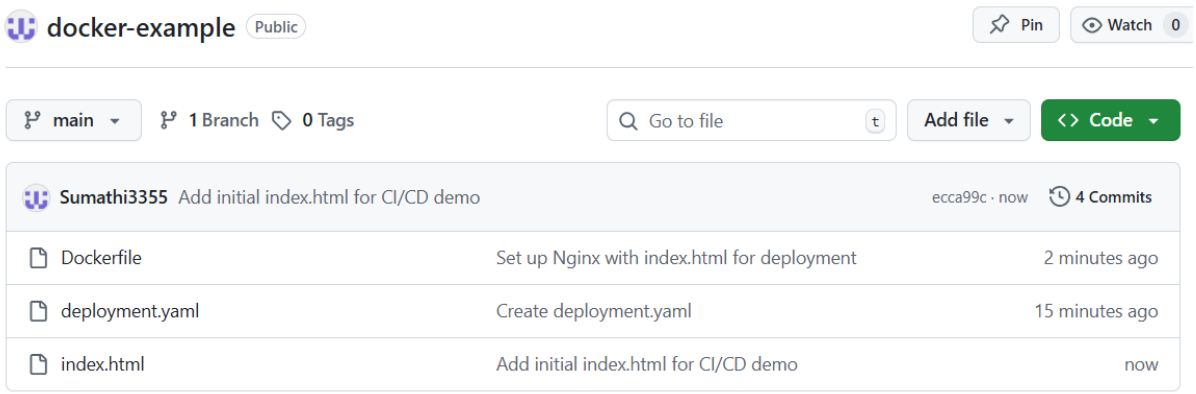
NAME : SUMATHI S

COURSE CODE : ISWE406P

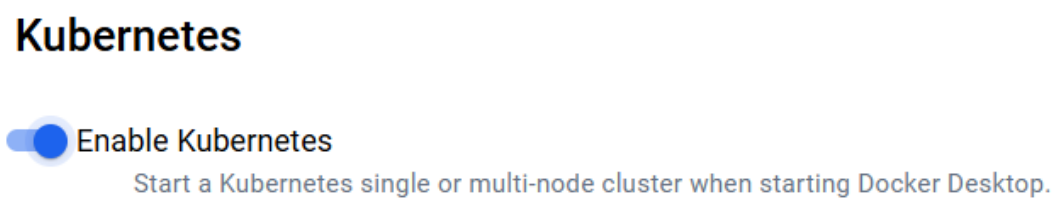
SLOT : L49+L50

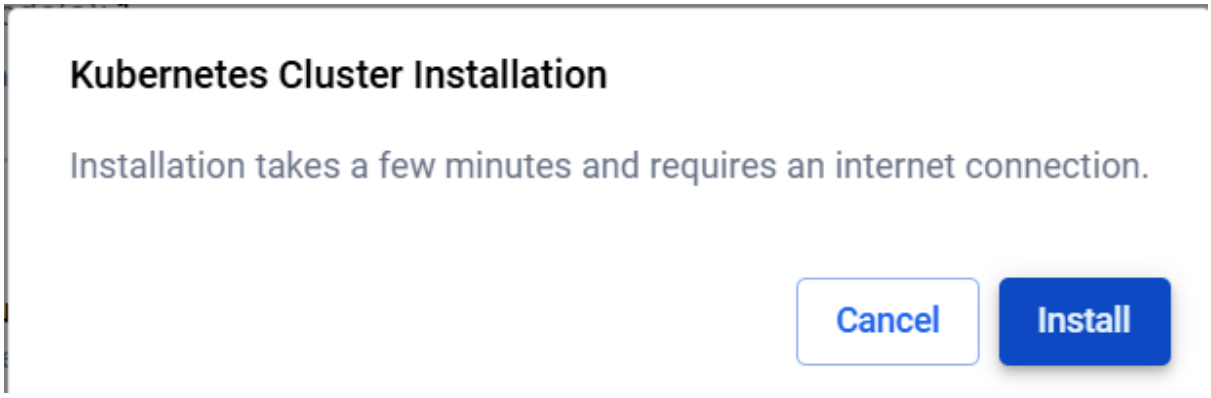
Create an HTML file for your profile and complete the following:

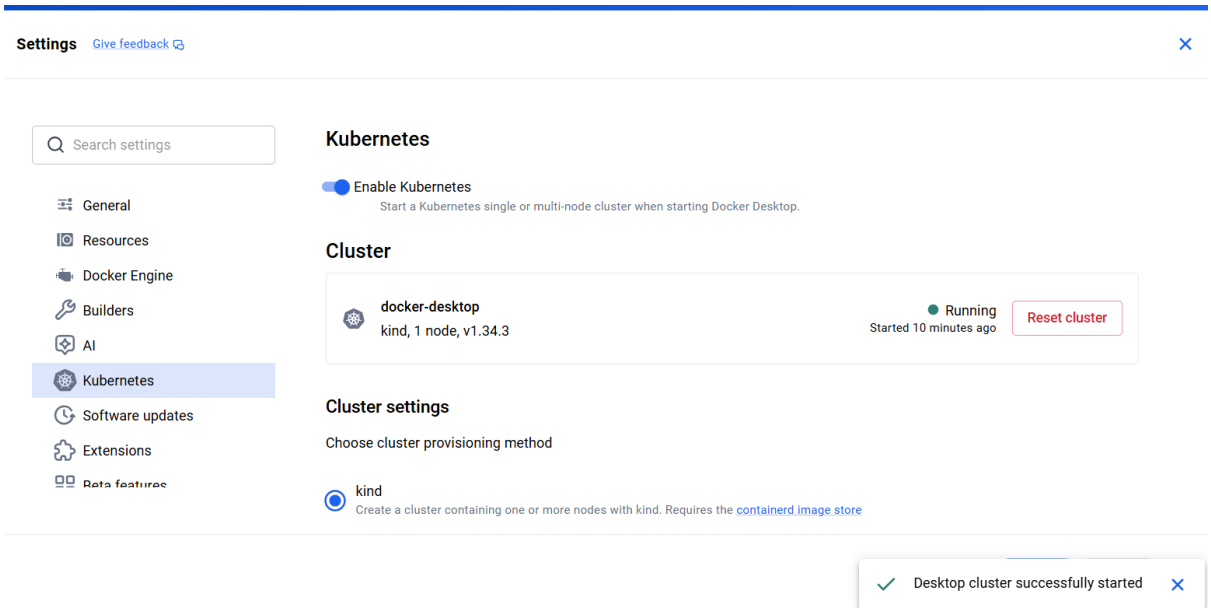
1. **Create a repository in GIT and create the following files:**
   - **index.html**
   - **Dockerfile**
   - **Deployment.yaml**



2. **Enable Kubernetes**
   - **Open the Docker Dashboard:** Launch the Docker Desktop application on your system.
   - **Navigate to Settings:** Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.
   - **Select the Kubernetes Tab:** In the Settings menu, find and select the Kubernetes tab.
   - **Enable Kubernetes:** Check the box labeled Enable Kubernetes.
   - **Apply & Restart:** Click the Apply & Restart button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time


Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.

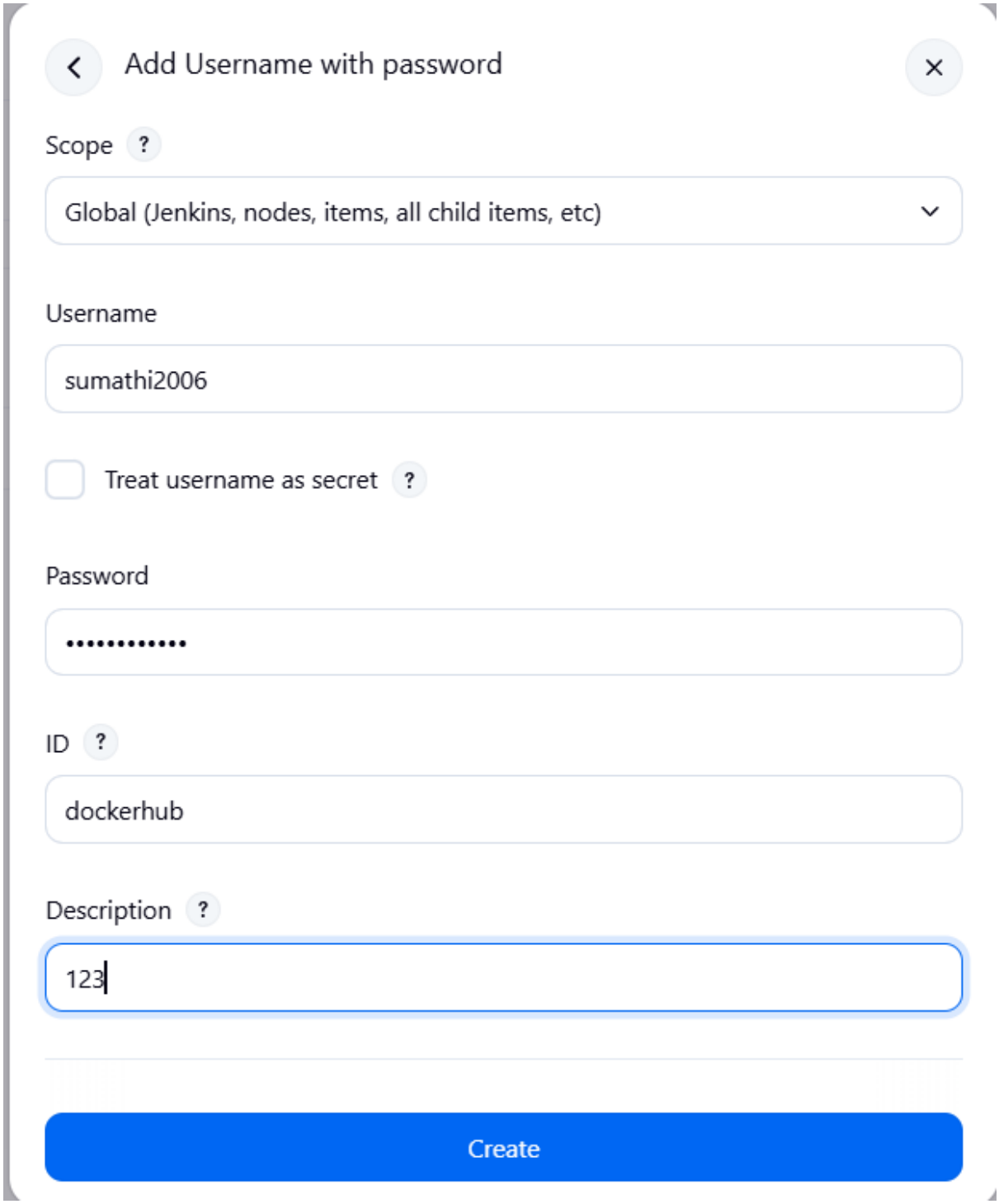

Kubernetes Cluster Installation
Installation takes a few minutes and requires an internet connection.
Cancel
Install


Settings
Give feedback
Search settings
General
Resources
Docker Engine
Builders
AI
Kubernetes
Software updates
Extensions
Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
Cluster
docker-desktop
kind, 1 node, v1.34.3
Running
Started 10 minutes ago
Reset cluster
Cluster settings
Choose cluster provisioning method
kind
Create a cluster containing one or more nodes with kind. Requires the containerd image store
Desktop cluster successfully started

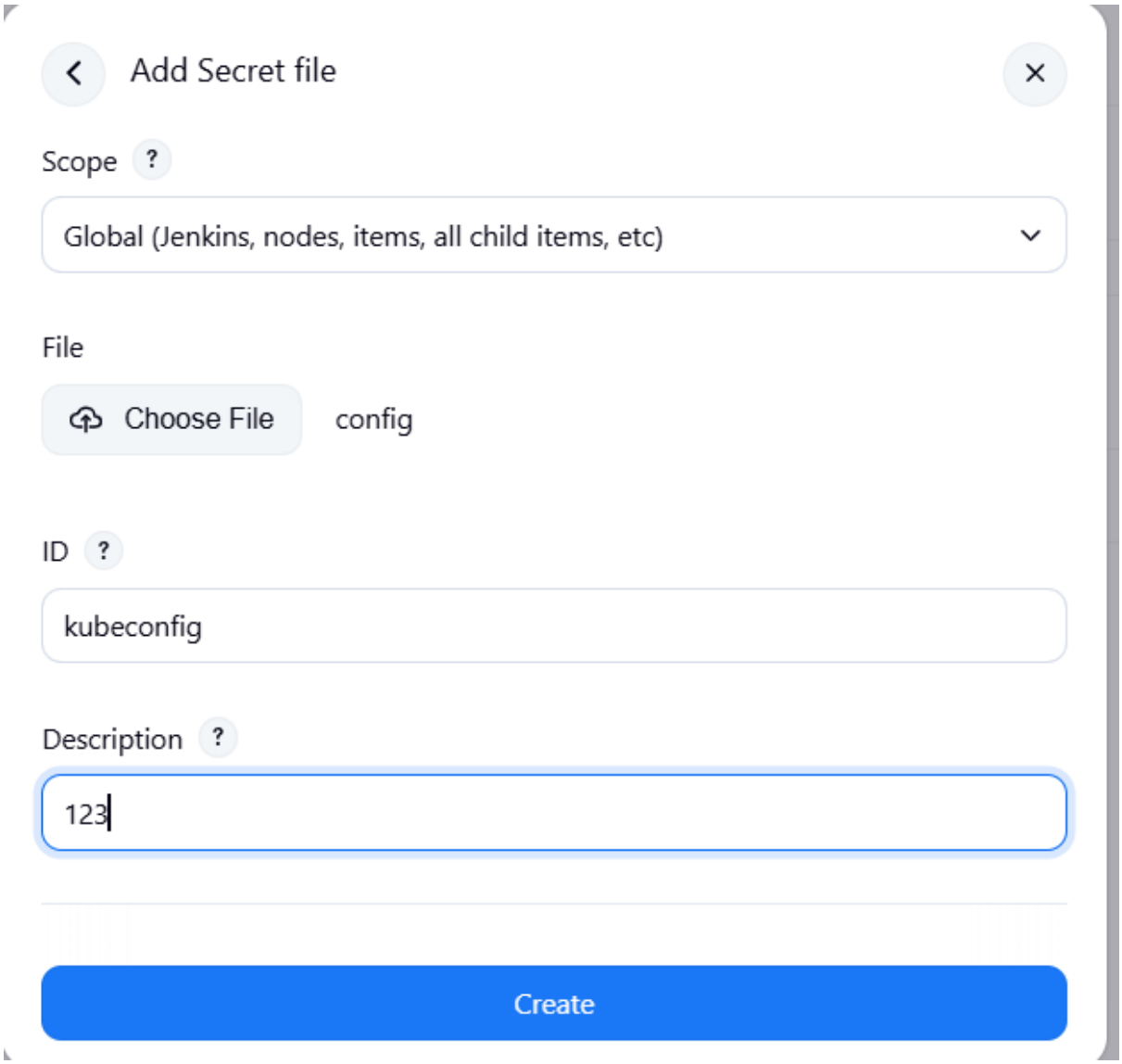
3. Open Jenkins

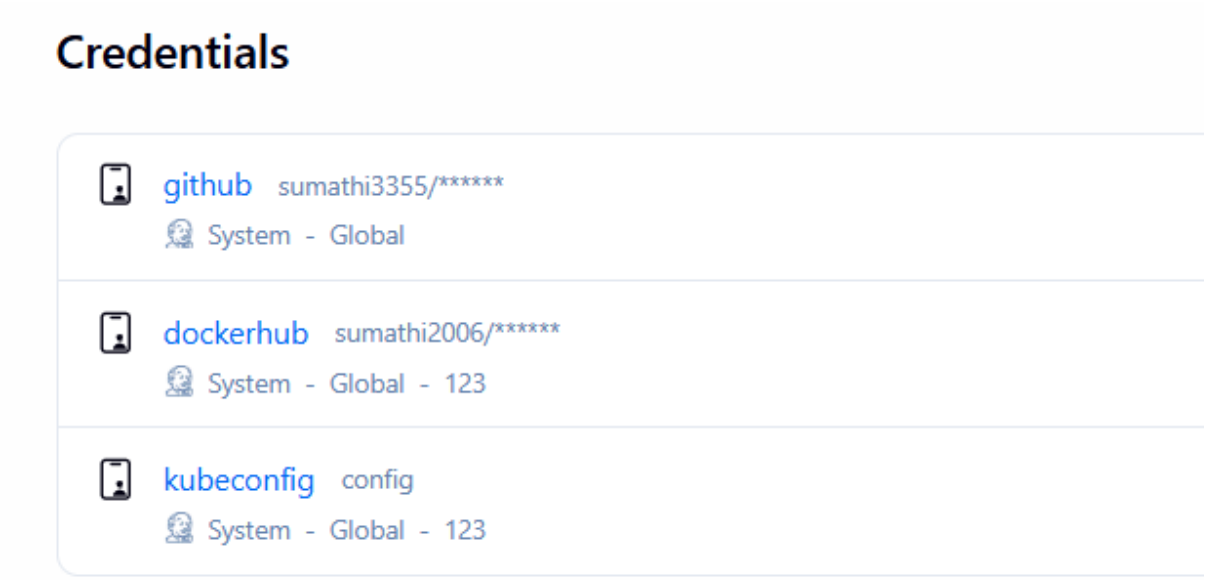
In Jenkins Setting --> Credentials

New Credentials – User name: <should use the Docker Desktop username>, password <same as Docker Desktop password>, ID:<dockerhub>, Description <any>



New Credentials – Secret – Broswse C: User->Administrator-->.kube





Create New Item -> Pipeline

Paste the code in pipeline script

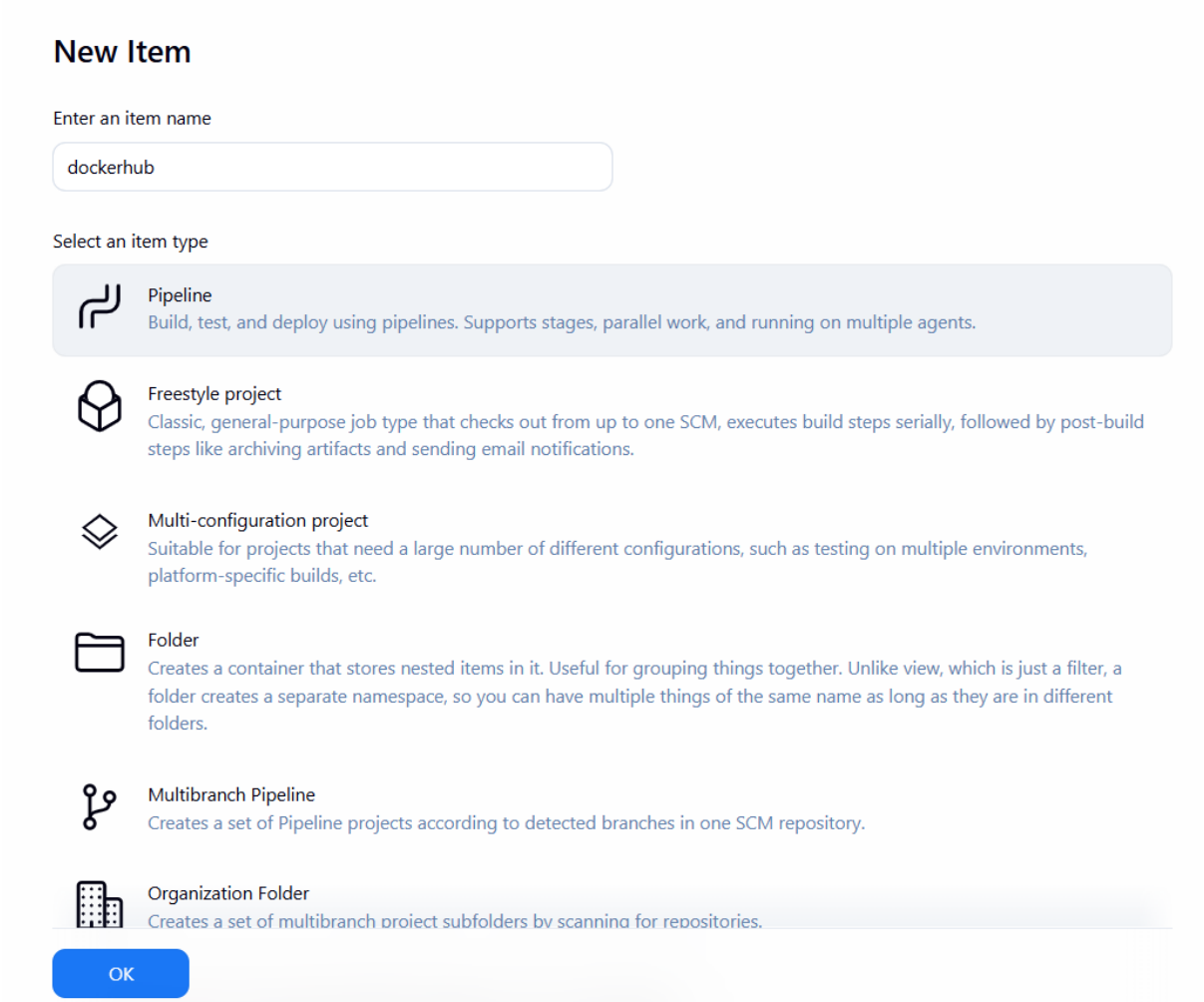
-Update URL of GIT repository

-Make sure credentialsId: “dockerhub” same as mentioned in credentials

-Make sure that the ID mentioned in credentials “kuberconfig” is same as in script

-Run the Build

Pipeline:


New Item
Enter an item name
dockerhub
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
OK

Pipeline script:

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

try sample Pipeline...

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "sumathi2006/html-demo"
    }

    stages {

        stage('Clone Code') {
            steps {
                git branch: 'main',
                url: 'https://github.com/Sumathi3355/docker-example.git'
            }
        }
```

Use Groovy Sandbox ?

Save Apply

Build now:

Jenkins / dockerhub

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

#11 7:39 PM

#10 7:32 PM

dockerhub

Stage View

	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~2min 42s)	14s	5s	51s	319ms
#11 Mar 26 19:39 No Changes	1s	3s	11s	835ms
#10 Mar 26 19:32 No Changes	1s	2s	4min 59s	1s
#9 Mar 26 19:30 No Changes	2min 2s aborted	145ms aborted	135ms aborted	141ms aborted
#8 Mar 26 19:29 No Changes	2s	3s	2min 44s	127ms

Jenkins / dockerhub / #11

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#11'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

#11 (Mar 26, 2026, 7:39:42 PM)

Started by user sumathi

This run spent:

- 12 ms waiting;
- 18 sec build duration;
- 18 sec total from scheduled to completion.

Revision: 71f167b7c9a10bb15d5d442add089808710d4cf2
Repository: https://github.com/Sumathi3355/docker-example.git

- refs/remotes/origin/main

No changes.

Console:

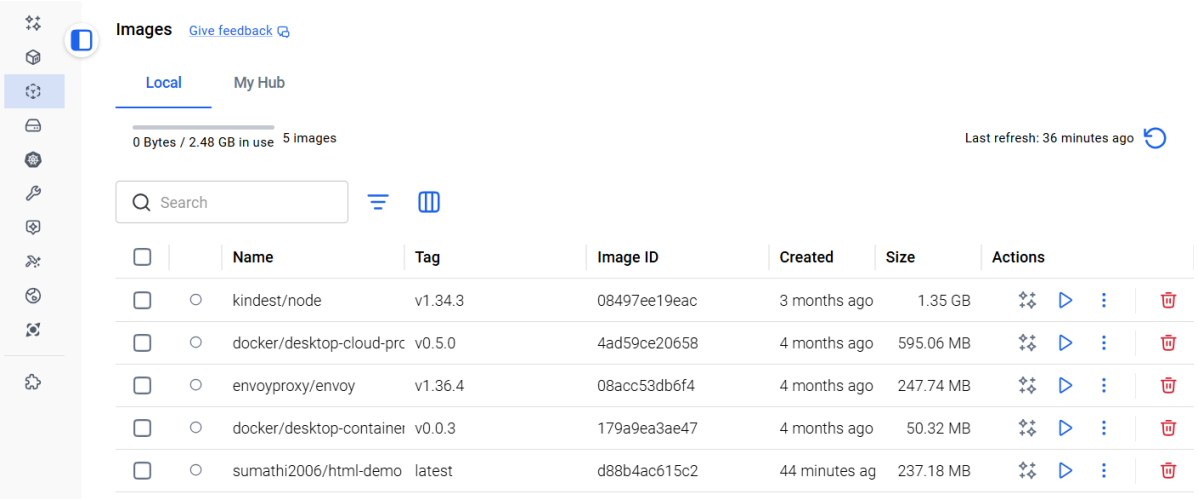
Console

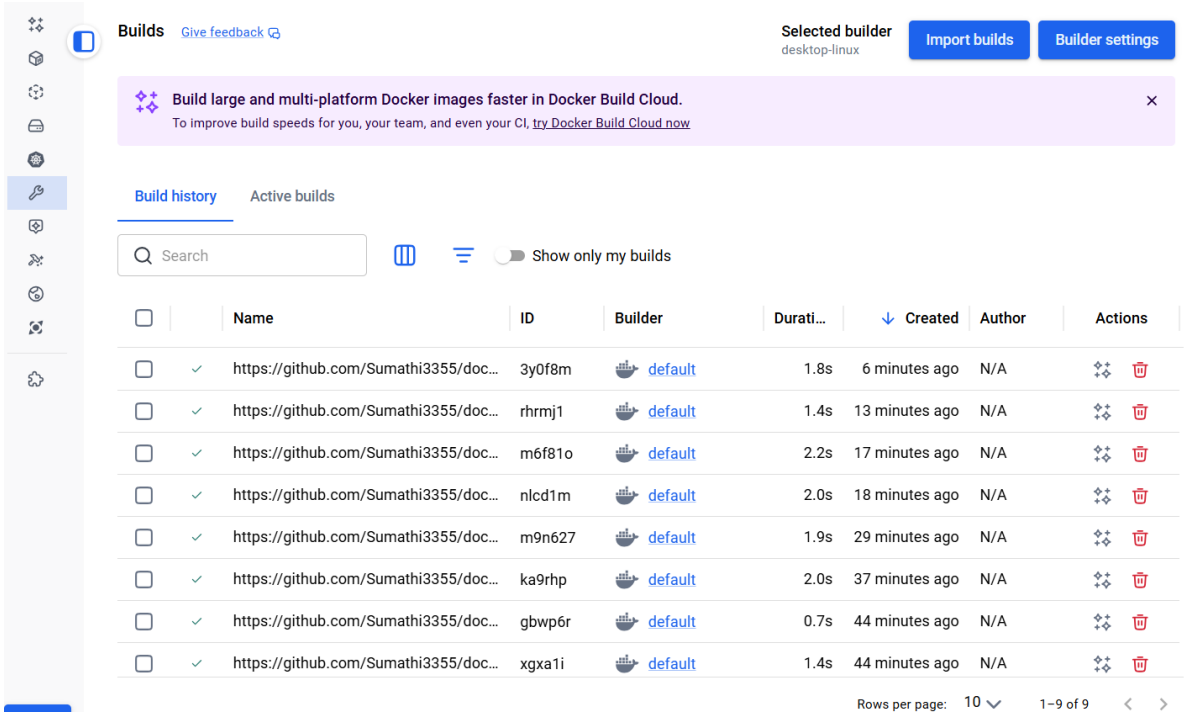
Download Copy View as plain text

```
Started by user sumathi
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\cmd\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub\.git # timeout=10
Fetching changes from the remote Git repository
 > C:\Program Files\Git\cmd\git.exe config remote.origin.url https://github.com/Sumathi3355/docker-example.git # timeout=10
Fetching upstream changes from https://github.com/Sumathi3355/docker-example.git
 > C:\Program Files\Git\cmd\git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > C:\Program Files\Git\cmd\git.exe fetch --tags --force --progress -- https://github.com/Sumathi3355/docker-example.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
 > C:\Program Files\Git\cmd\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 71f167b7c9a10bb15d5d442add089808710d4cf2 (refs/remotes/origin/main)
 > C:\Program Files\Git\cmd\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -f 71f167b7c9a10bb15d5d442add089808710d4cf2 # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -a -v --no-abbrev # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -D main # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -b main 71f167b7c9a10bb15d5d442add089808710d4cf2 # timeout=10
Commit message: "Update Docker image for html-demo deployment"
 > C:\Program Files\Git\cmd\git.exe rev-list --no-walk 71f167b7c9a10bb15d5d442add089808710d4cf2 # timeout=10
```

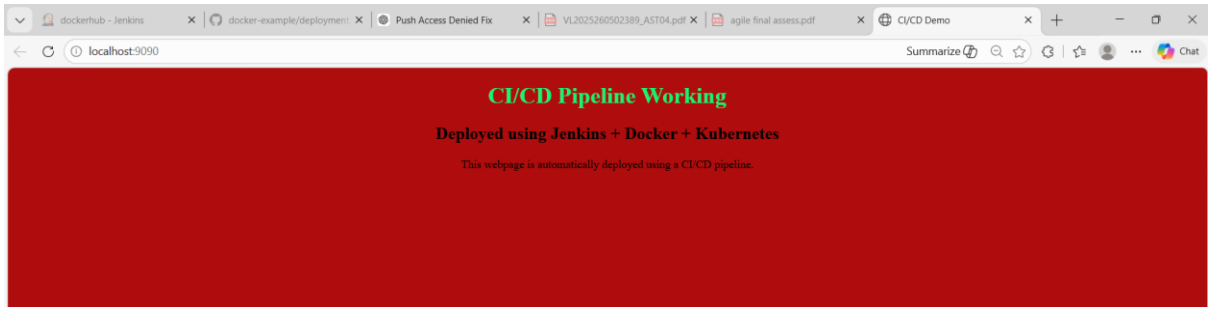
```
bc29762bf937: Waiting
bc29762bf937: Waiting
bc29762bf937: Pushed
latest: digest: sha256:d88b4ac615c2edeccf6cc20d010cad60823f1a14771284b942ba88385edc08f1 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub>set KUBECONFIG=****

C:\ProgramData\Jenkins\.jenkins\workspace\dockerhub>kubectl apply -f deployment.yaml --validate=false
deployment.apps/html-demo unchanged
service/html-demo-service unchanged
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

*** The same should be reflected in Dockerdesktop





Also the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010

So check if the HTML file open in http://localhost:30010

```
PS C:\Users\ssuma> kubectl port-forward service/html-demo-service 9090:80
Forwarding from 127.0.0.1:9090 -> 80
Forwarding from [::1]:9090 -> 80
Handling connection for 9090
Handling connection for 9090
```